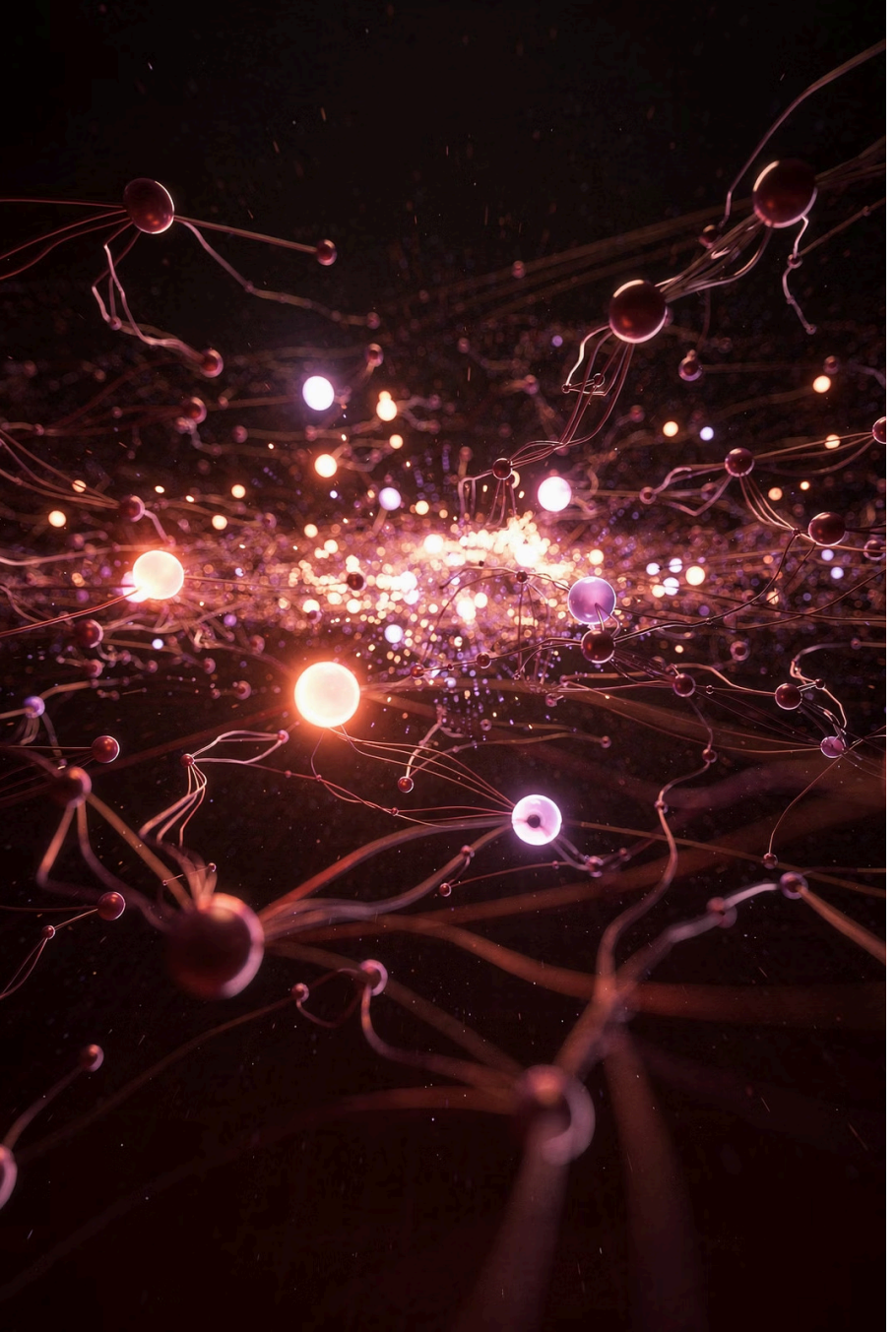


תל-חי
אוניברסיטת
קריית שמונה בגליל

Introduction to Data Science

Welcome & Overview

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LECTURE 1



What Is Data Science?

- Interdisciplinary field that turns data into insights
- Uses scientific methods, algorithms, and systems to support decisions

Real-World Applications of Data Science

Data science is transforming nearly every sector, creating innovative products and services that drive efficiency and personalized experiences.

Healthcare Diagnostics

AI-powered tools assist in early disease detection, drug discovery, and personalized treatment plans.

E-commerce Recommendations

Sophisticated algorithms suggest products and content, enhancing user experience and sales for online platforms.

Financial Fraud Detection

Machine learning models identify suspicious transactions in real-time, preventing financial crime and securing assets.

Autonomous Vehicles

Data science is critical for processing sensor data, enabling navigation, obstacle detection, and predictive maintenance in self-driving cars.



Key Skills of a Data Scientist

Data science is a multidisciplinary field, requiring a blend of technical and soft skills to extract meaningful insights from data and drive informed decision-making.



Programming Proficiency

Mastery of languages like Python and R for data manipulation, statistical analysis, and model building.



Statistical & Mathematical Foundations

Strong understanding of statistics, probability, linear algebra, and calculus for robust analysis.



Machine Learning Expertise

Knowledge of various ML algorithms, from supervised and unsupervised learning to deep learning techniques.



Data Wrangling & Preprocessing

Skills in cleaning, transforming, and preparing raw, messy data for analysis and model training.



Communication & Storytelling

Ability to translate complex findings into clear, actionable insights and present them effectively to diverse audiences.



Domain & Business Acumen

Understanding the specific industry or business context to ask the right questions and interpret results accurately.

Problem Types Solved by Data Science

Data Science tackles a diverse array of challenges by applying specialized analytical techniques. Understanding these problem types helps frame how data can be used to derive actionable insights.



Classification

Categorizing data into predefined groups or labels, such as spam detection, disease diagnosis, or image recognition.



Regression

Predicting continuous numerical values based on input data, like forecasting house prices, stock market trends, or temperature.



Clustering

Grouping similar data points together without prior labels, often used for market segmentation or document analysis.



Anomaly Detection

Identifying rare occurrences or outliers that deviate significantly from expected patterns, critical for fraud detection or system monitoring.



Time Series Analysis

Analyzing data points collected over time to understand past behavior, forecast future values, or identify periodic trends.



Advanced Applications

Encompassing more specialized areas like Natural Language Processing (NLP), Computer Vision, and sophisticated Recommendation Systems.

CHAPTER 2 THE LIFECYCLE

The Data Science Lifecycle

Problem Formulation

Define the question clearly before touching any data

Data Collection

Gather relevant data from databases, APIs, surveys, or sensors

Cleaning & Preprocessing

Handle missing values, remove noise, and fix inconsistencies

Exploratory Data Analysis

Visualize distributions, find correlations, spot anomalies

Modeling & Evaluation

Build predictive models, validate with metrics

Deployment & Monitoring

Put models into production, track performance over time

Results Dissemination

Present findings clearly to stakeholders through reports, dashboards, and storytelling

CHAPTER 2 | PROBLEM FORMULATION

Problem Formulation

Good vs. Bad Research Questions

✅ Good Research Question

A strong research question is **S.M.A.R.T.** (Specific, Measurable, Achievable, Relevant, Time-bound). It must be clear, focused, and answerable using available data, driving actionable insights.

Example: *"How does customer engagement with our mobile app features (e.g., in-app purchases, daily active sessions) correlate with 3-month churn rates for users acquired in Q1 2023?"*

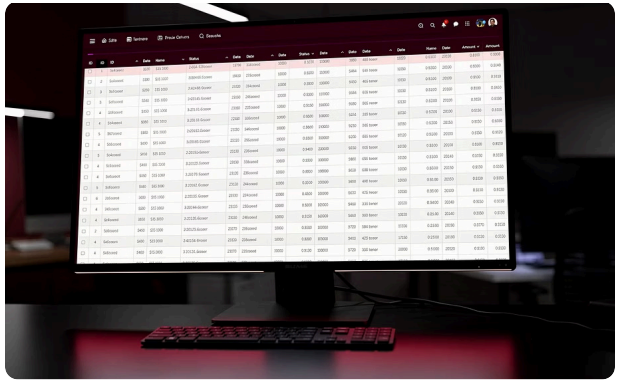
❌ Bad Research Question

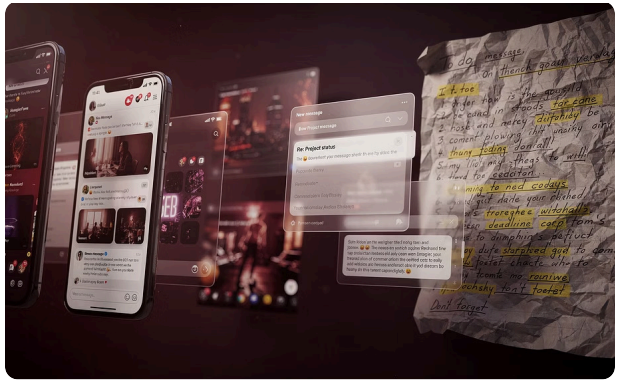
Poor questions are often vague, subjective, too broad, or not empirically testable. They lead to ambiguous findings or cannot be addressed with data science methods.

Example: *"Are our customers happy with the new app update?"* (Too vague, "happy" is subjective without metrics.)

Data Collection

Structured vs Unstructured Data





Structured Data

Organized in rows and columns — relational databases, spreadsheets, CSV files. Easy to query with SQL.

Unstructured Data

No predefined format — emails, images, audio, video, social media posts. Requires NLP or computer vision to process.

Approximately **80–90%** of the world's data is unstructured, making it one of the biggest challenges in modern data science.



Data Collection

Introduction to APIs in Data Science

APIs (Application Programming Interfaces) are gateways that let your code request data directly from external services — weather stations, social platforms, financial markets, and more.



🧪 **Mini Exercise:** Which of these are accessible via APIs? (a) Real-time stock prices (b) Satellite imagery (c) Public transit schedules (d) All of the above → **Answer: d!**

Types of Variables
Quantitative Variables



Continuous

Any value within a range — height (172.3 cm), temperature (36.7°F), time elapsed. Measured on an infinite scale.



Discrete

Countable whole numbers — number of visits (3), items purchased (12), rooms in a house. No fractional values.

Types of Variables
Qualitative Variables

Nominal

Categories **without** inherent order

Examples: colors, gender, city names

Ordinal

Categories **with** meaningful order

Examples: satisfaction (low → high), education level

🧪 **Mini Exercise:** Classify each survey variable:
1. Favorite cuisine → 2. Income bracket (low/mid/high) → 3. Zip code → 4. Star rating (1–5) →

Real-World Data Challenges

Missing Data

Incomplete records create gaps in analysis

Noise

Random errors and irrelevant information obscure signals

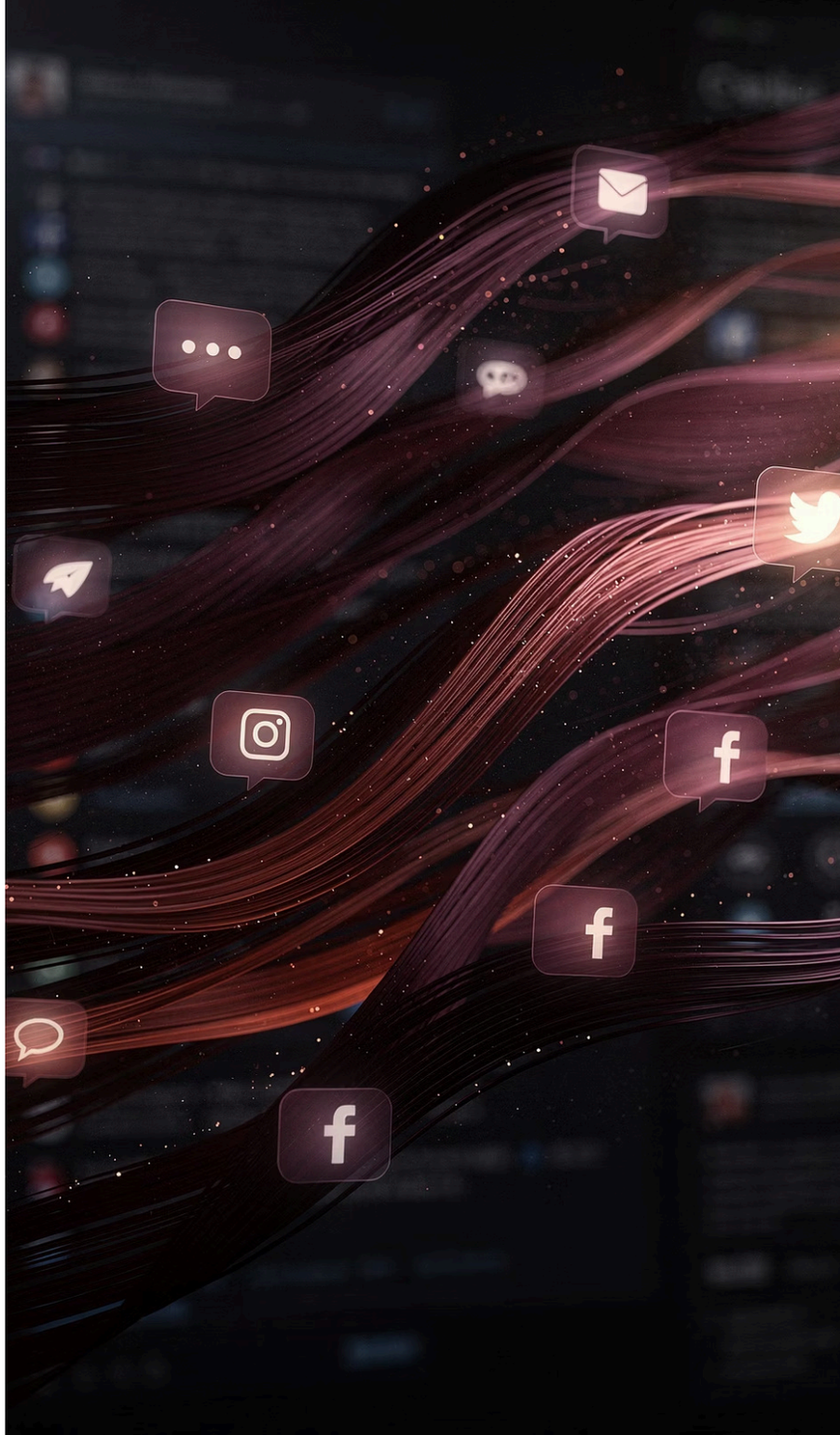
Bias

Systematic skew in collection leads to unfair outcomes

Volume & Variety

Massive scale and diverse formats demand robust pipelines

💬 **Discussion:** Imagine analyzing Twitter data for sentiment analysis. What challenges might arise? Think about sarcasm, slang, bots, and multilingual content.



Data Cleaning Strategies

1

Imputation

Fill missing values using mean, median, or predictive methods

2

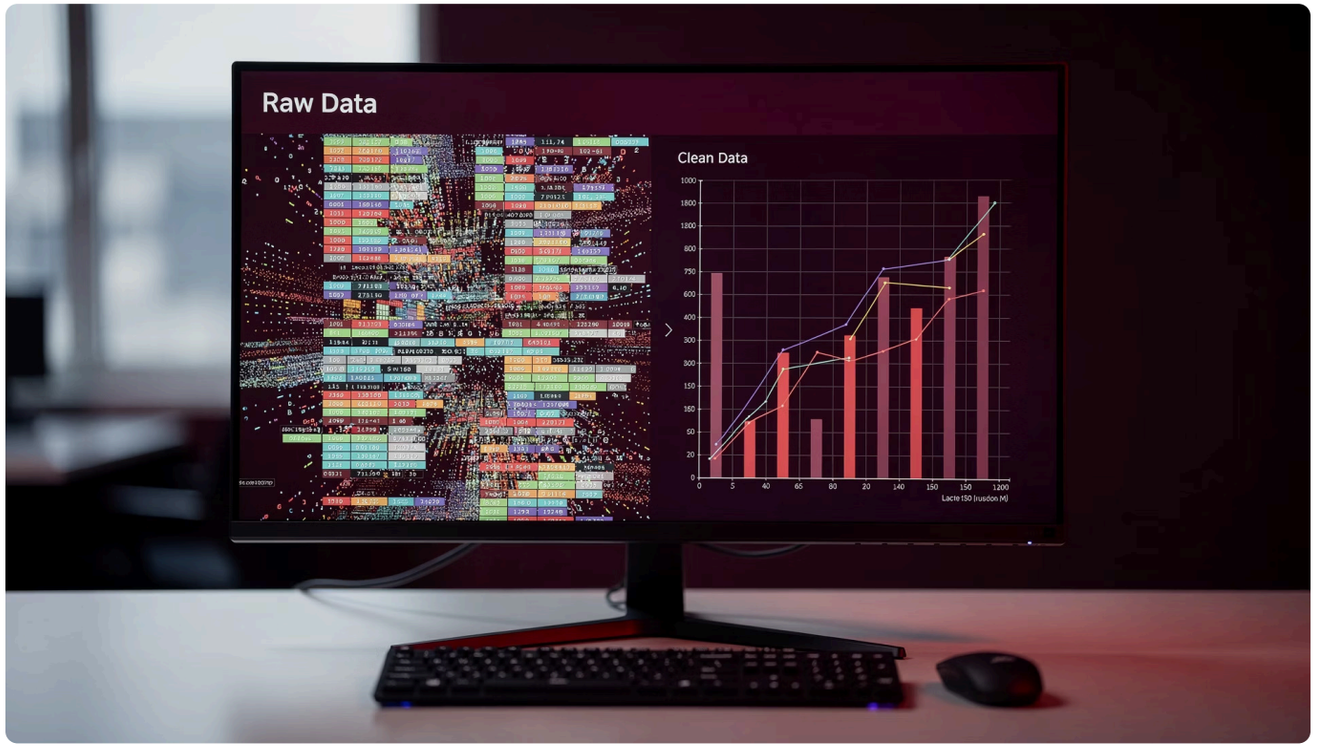
Outlier Detection

Identify and handle extreme values using IQR or z-scores

3

Normalization

Scale features to comparable ranges for fair modeling



Ethics in Data Gathering: Why It Matters

Privacy

Protect individuals' personal information

Consent

Ensure participants knowingly agree to data use

Bias

Recognize and mitigate systematic unfairness

Transparency

Be open about methods, sources, and limitations

💬 **Case Study Prompt:** A hospital wants to use patient health records for AI research. What ethical issues should they consider before proceeding?



Ethical Data Practices

Anonymization

Remove or mask personally identifiable information before analysis

Data Minimization

Collect only what is strictly necessary for the stated purpose

Informed Consent

Clearly explain how data will be used and obtain explicit agreement



💬 **Discussion:** "How can bias in training data affect outcomes?" Consider hiring algorithms trained on historically biased résumé data.

Summary: Key Takeaways

Data Types & Variables

Understanding quantitative vs. qualitative and structured vs. unstructured data is foundational to every analysis

The Lifecycle

Data science follows a rigorous process — from problem formulation through deployment and monitoring

Ethics

Responsible data gathering and ethical awareness are integral to trustworthy data science



Course Technicalities

1

Schedule

Lectures & Labs: Thursday

2

Grading Components

- 4 Homework Assignments: 40% (pairs)
- Final Project: 45% (pairs)
- Class Assignments: 15% (alone)

3

Key Requirements

- Submit 70% of class exercises by 21:00 on class day (only if you were present during the class!)
- Participate in 70% of classes

COURSE DETAILS | EVALUATION

Interactive Evaluation Components

Your learning and understanding will be assessed through engaging discussions and presentations:

1

Final Project Presentation

~15-minute presentation and direct discussion showcasing your work.

2

Lab Discussions

A portion of lab grade determined through one-on-one discussions about submitted lab assignments.

3

Homework Evaluation

Partially evaluated based on direct discussions of solutions and approach.

Q&A and Open Discussion

> "The goal is to turn data into information, and information into insight."
>
> — Carly Fiorina

This is your time — ask about anything we covered today. No question is too basic or too advanced.

💬 **Prompt to get started:** "What excites or concerns you most about data science? Where do you see it impacting your field of study?"





Thank You!

Go explore data — get your hands dirty, break things, and learn by doing.

📧 Email

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🕐 Office Hours

Per Request

📍 Location

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